ModelIt!

Student Activity Pack

The Ocean's Silent Emergency

Modeling Ocean Acidification, pH Tipping Points, and Marine Ecosystem Collapse

Grade 11L2 | NGSS HS-ESS2-6, HS-LS2-6 | Stability and Change / Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think happens to the ocean when it absorbs carbon dioxide from the atmosphere?

2 Why might a small change in ocean chemistry matter to organisms that build shells?

3 Draw a diagram showing how you think CO2 moves from the atmosphere into the ocean and what it does there.

4 If the ocean is becoming more acidic, why don't we hear about it as much as we hear about climate change?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Atmospheric CO2 Concentration
- Ocean CO2 Absorption Rate
- Seawater pH
- Carbonate Ion Availability
- Shell-Building Organism Survival
- Marine Food Web Stability

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Pre-Industrial Baseline

CO_2 at 280 ppm, pH at 8.2, stable carbonate levels

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Current Trajectory

CO_2 rising to 900+ ppm, pH dropping to 7.7

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Aggressive Mitigation

CO_2 stabilized at 450 ppm, pH decline slowed

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Ocean Temperature — Rising sea surface temperatures reduce CO_2 solubility but increase metabolic stress on marine organisms, creating a complex interaction with acidification effects

Dissolved Oxygen Level — Warmer, more acidic water holds less dissolved oxygen, creating hypoxic zones that compound the stress on marine organisms already struggling with acidification

Phytoplankton Productivity — Microscopic ocean plants that produce 50% of Earth's oxygen and form the base of marine food webs — some species are enhanced by higher CO_2 while others are harmed by lower pH

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What happens to the ocean — and everything living in it — when seawater absorbs too much carbon dioxide? Use evidence from your simulation to support your answers.

1. **Why is the rate of current ocean acidification more dangerous than past natural pH changes, even though the ocean has experienced lower pH in its geological history?**

2. **How does your model reveal that ocean acidification is a 'hidden' climate crisis compared to warming temperatures?**

3. **What feedback loops in your model could make acidification self-reinforcing once certain tipping points are crossed?**

4. **How would the collapse of shell-building organisms affect human communities that depend on fisheries?**

5. **What does your model suggest about the relationship between the speed of CO_2 emission reductions and the ocean's ability to recover?**

8 STEM Challenge: Design an Ocean Acidification Early Warning System

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YOUR ENGINEERING MISSION

Design a monitoring and response system that can detect dangerous pH changes in a marine ecosystem and trigger protective interventions before tipping points are crossed.

THE SCENARIO

A coastal marine sanctuary has noticed declining shellfish populations and wants to understand whether ocean acidification is approaching a critical threshold. Your team has been contracted to design a comprehensive monitoring network and adaptive management plan that uses real-time pH data to guide conservation decisions.

Engineering Design Thinking

1. WHAT TYPES OF SENSORS AND HOW MANY MONITORING STATIONS WOULD YOUR NETWORK REQUIRE?
2. HOW WOULD YOU INTEGRATE PH DATA WITH BIOLOGICAL SURVEYS TO CONFIRM ECOSYSTEM IMPACTS?
3. WHAT LOCAL ALKALINITY ENHANCEMENT STRATEGIES COULD TEMPORARILY BUFFER CRITICAL HABITATS?
4. HOW WOULD YOU COMMUNICATE ACIDIFICATION RISK TO FISHERIES, TOURISM OPERATORS, AND POLICYMAKERS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Chemical Oceanographers study the chemistry of seawater and how it affects marine life, climate, and global carbon cycles. They deploy sensors, analyze water samples, and build predictive models for organizations like NOAA, Scripps Institution of Oceanography, and Woods Hole Oceanographic Institution, earning $75,000–$140,000/year. Marine Conservation Scientists designing acidification adaptation strategies earn $65,000–$120,000/year.